Multilevel Caches

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Local and global miss rates

Local miss-rate: Number of misses in a cache at level i divided by the total number of accesses at level i.
Global miss-rate: Number of misses in a cache at level i divided by the total number of accesses to memory hierarchy.

Example: A core sends 10,000 accesses to L1, of which, 9,000 hit. Remaining 1,000 go to L2, of which, 600 hit.

	Local miss-rate	Global miss-rate
L1 cache	1000/10,000	1000/10,000
L2 cache	400/1,000	400/10,000

For L1 cache, local and global miss-rate are same. If nothing is mentioned, it is local miss-rate

Multilevel Caches

- L1 cache attached to CPU
  - Small, but fast
- L2 cache services misses from L1 cache
  - Larger, slower, but still faster than main memory
- Some systems include L3 and L4 cache
- Main memory services L2 cache misses (assuming no L3/L4)

Multilevel Cache Example 1 [1 of 2]

- Given
  - CPU base CPI = 1, clock rate = 4GHz
  - Miss rate/instruction of L1 I-cache = 2%
  - Main memory access time = 100ns
- Find CPI with realistic caches.
- Solution: With just L1 I-cache, misses will go to main memory
  - Miss penalty = 100ns/0.25ns = 400 cycles
  - Effective CPI = $1 + 0.02 \times 400 =$ **9**

Multilevel Cache Example 1 [2 of 2]

- Now add L-2 cache
  - Access time = 5ns
  - Global miss rate of L2 cache= 0.5%. Find new CPI
- Case 1: L1 miss with L2 hit
  - Penalty = 5ns/0.25ns = 20 cycles
- Case 2: L1 miss with L2 miss
  - Extra penalty = 400 cycles
- CPI = $1 + 0.02 \times 20 + 0.005 \times 400 =$ **3.4**
- Performance ratio = 9/3.4 = 2.6

Multilevel Cache Example 2

We will now solve the previous question with just one change:

~~Global~~ Local miss rate of L2 cache= 0.5%.

For every 1 instruction, 0.02 missed from L1 I-cache. They go to L2 cache. Of these 0.02 accesses to L2 cache, 0.02*0.005 miss from L2 cache and they go to main memory

Case 1: L1 miss with L2 hit. Penalty =20 cycles

Case 2: L1 miss with L2 miss. Penalty = 400 cycles

CPI = 1 + (0.02 × 20) + (0.02 × 0.005 × 400) = **1.44**

Multilevel Cache Considerations

- L1 cache
  - Focus on minimal hit time
- L2 cache
  - Focus on low miss rate to avoid main memory access
  - Hit time has less overall impact
- Results
  - L1 block size is equal or smaller than L2 block size.
  - Reason: increasing L1 block size would increase hit time
  - Increasing L2 block size would reduce L2 miss-rate

L2 cache: hit-time or miss-rate

- We just did this calculation:
- CPI = 1 + 0.02 × 20 + 0.005 × 400 = **3.4**
- **Now, double the hit-time to 40 cycle**
- CPI = 1 + 0.02 × 40 + 0.005 × 400 = **3.8**
- **Instead, double the miss-rate to 1%**
- CPI = 1 + 0.02 × 20 + 0.010 × 400 = **5.4**
- **Lesson: In L2 cache, hit-time increase has less impact than miss-rate increase.**